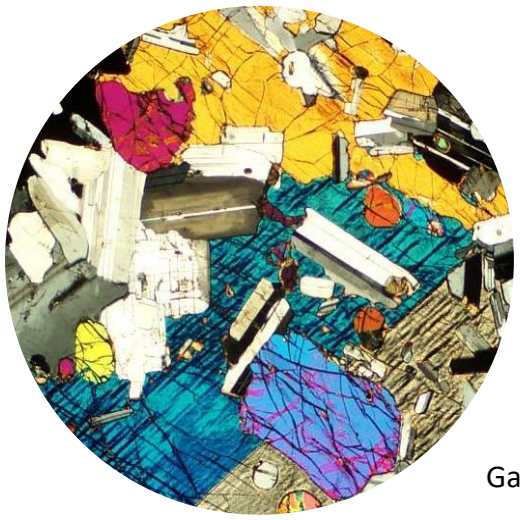
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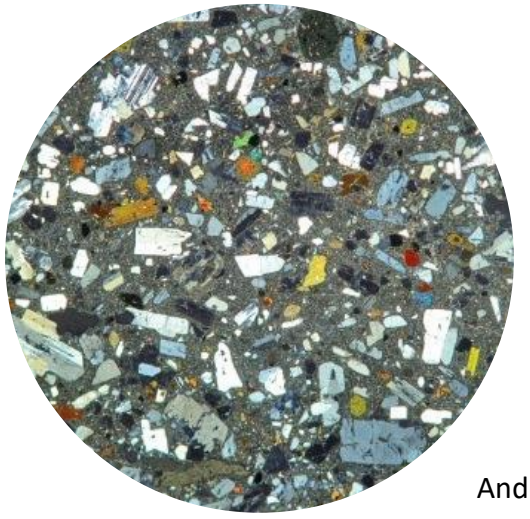
EART22201

Igneous Minerals and Processes

Practical Reference Handout



Gabbro, Skaergaard



Andesite, Japan

This booklet is a guide to help you with EART22201 Igneous Petrology practicals.
You should use it in addition to the recommended course text books.

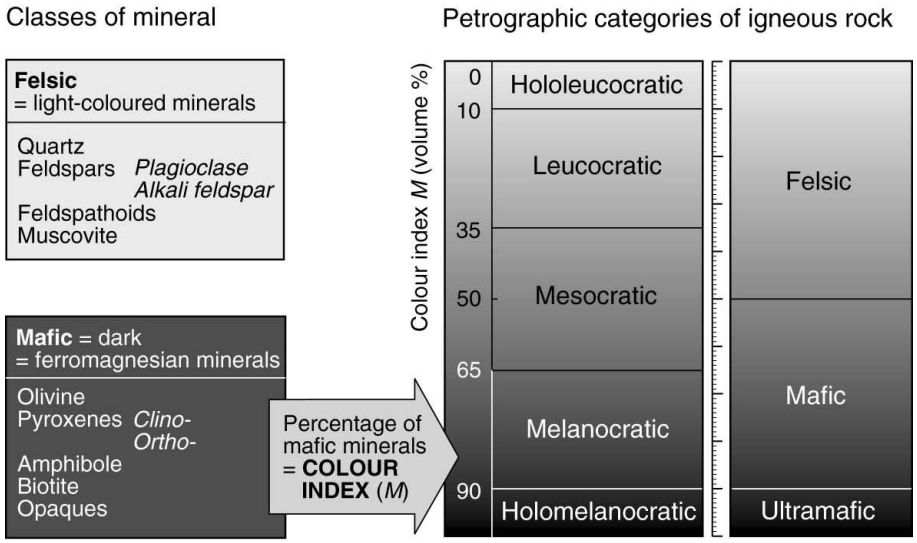
Some information and figures in this handout have been taken from
Gill (2010): *Igneous rocks and processes: A practical guide*, downloaded from the e-book.

Please bring this handout to <u>all</u> practical classes

A guide to the systematic description of igneous rocks

Hand Specimens

1. Colour



Gill (2010) Fig. 1.3b

2. Crystallinity and grain size

Average mineral diameter	Textural terms
>30 mm	pegmatitic
5-30 mm	very coarse-grained
3-5 mm	coarse-grained (groundmass crystals can be identified with the naked eye)
1-3 mm	medium-grained (groundmass crystals are visible to the naked eye but too small to identify)
0.05-1 mm	fine-grained (groundmass crystals are too small to see with the naked eye)
<0.05 mm	very fine-grained

It is best to report a measurement of the average mineral diameter alongside a qualitative textural description of coarse-, medium- or fine-grained.

3. Presence or absence of phenocrysts

The texture is porphyritic if phenocrysts are present: use "XX-phyric" for a rock with phenocrysts of mineral XX. If the rock has no phenocrysts, use even-grained if the rock is crystalline, or aphyric if the rock is a lava or is glassy.

> The above three descriptions can usually be written down in a single sentence. Example: "This melanocratic rock is hypocrystalline, very fine-grained, and is olivine-phyric."

4. Structure: describe any banding, lineation, jointing, vesicularity, and anything else of interest.

5. Identify all the minerals present as far as possible. Note any special features of the minerals (e.g. obvious cleavage). Note the state of alteration of the rock and the minerals.

When describing outcrops or hand specimens in the field, remember that a weathered surface may be just as informative as the freshly broken surface.

Thin Sections

1. Describe the features visible to the naked eye: grain size, presence or absence of phenocrysts, any layering or banding, any contacts or inclusions.

2. Examine the thin section under the microscope. Before writing or drawing anything, look around the whole section to make sure you have seen all the different minerals and an overview of the rock texture.

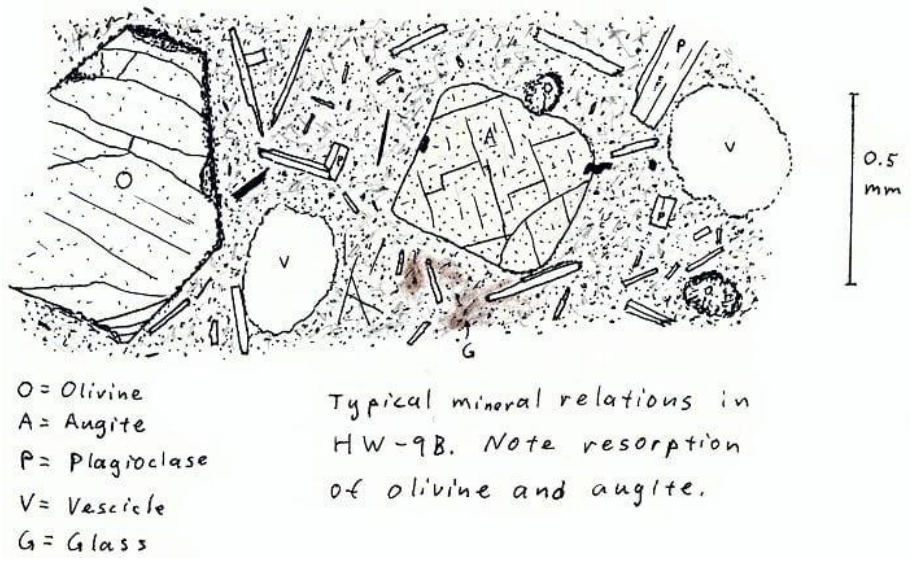
3. Drawing a good thin section diagram

A thin section diagram is often referred to as a "sketch", but in reality it is a technical drawing that captures a representative view of the rock, its mineralogy and its key textural features.

- Make the sketch large. Give yourself at least half a page to sketch and add labels. A good size for the sketch would be 10 cm.
- Draw at an appropriate scale for the grain size of the rock. You may need to do several drawings at different scales. If the rock is very coarse grained, you will need to draw a zoomed-out view to convey the texture.
- Include a scale bar. You can estimate the field of view by putting a transparent ruler under the microscope. Don't use a "x10" or "x20" label – this only gives information about the objective lens you used to make the observations; it doesn't translate to the scale of your diagram.
- Use line weight, shading and colour to add detail.
- The sketch should show all the different minerals in the rock, and all the textural features that you mention in your description.

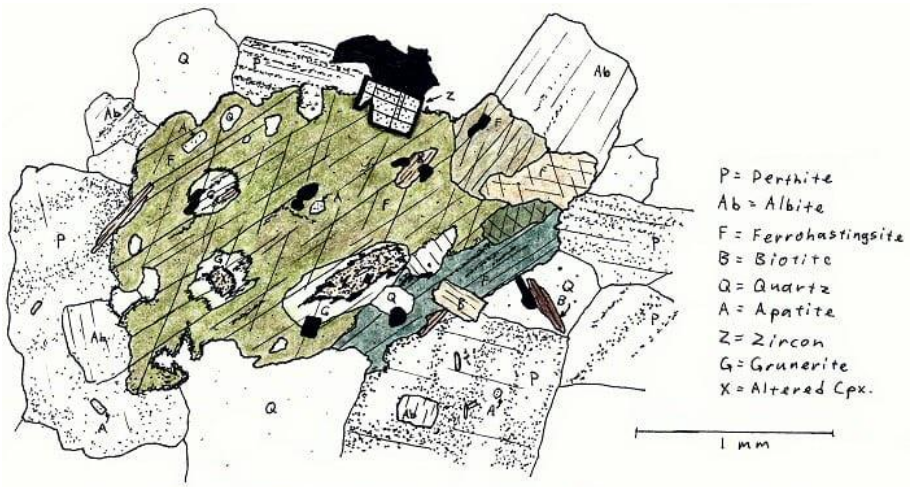
Formal reports such as dissertations often present photographs of thin sections instead of hand-drawn sketches. The author needs to choose whether plane-polarized light or crossed polars (or both) will best show the important features of the rock. Photographs must have a scale bar and an appropriate caption, and can include unobtrusive labels.

Here are a couple of good examples of thin section sketches produced by a student in a petrology class (see version on Blackboard for colour diagrams). More here: https://tinyurl.com/yn9z622a



O = Olivine
A = Augite
P = Plagioclase
V = Vesicle
G = Glass

Typical mineral relations in HW-9B. Note resorption of olivine and augite.



Typical mineral relations. Note alteration of remnant clinopyroxene to grunerite within ferrohastingsite.

4. Writing a good thin section description

The thin section description contains both observations and interpretation. It concisely summarizes the observational information contained in the sketch, adding details where needed. It then uses the observational evidence to interpret the sequence of events that formed the rock.

a) Describe the individual minerals present. Describe the minerals in the following order:

- Primary minerals – large crystals (e.g. phenocrysts, cumulus crystals)
- Primary minerals – small crystals (e.g. groundmass, interstitial minerals)
- Accessory minerals, e.g. spinel, opaque oxides, apatite, sphene, zircon
- Secondary minerals, e.g. serpentine, chlorite

The mineral description should include some or all of the following information:

- Size, habit and shape of the grains
- Colour and pleochroism
- Composition (only if it is possible to obtain this information from the thin section)
- Any twinning, zoning or exsolution
- Any alteration (note which mineral has been altered, e.g. olivine to serpentine)

You may find it useful to arrange the information in a list or table.

b) Describe the texture

The texture is the spatial and geometric relationship between the crystals. Describe features such as:

- A preferred mineral alignment or banding
- One mineral included within another
- The range of crystal sizes. Is one mineral noticeably larger or smaller than the others? Are any minerals present across a range of crystal sizes?
- Is there any evidence of deformation or compaction, e.g. bent twin planes, undulose extinction?

The key textural features should be captured and labelled in the sketch.

Use the texture **determine the order of crystallization** as far as possible. In most igneous systems, multiple minerals crystallize together simultaneously, so Bowen's Reaction Series is not usually good evidence for the order of crystallization. Instead, look at the rock's texture to try to distinguish 'earlier' from 'later' phases. Textural evidence to support the proposed order of crystallization might include:

- When one mineral has grown around another, the surrounded mineral is older and the enclosing mineral is younger.
- Early-formed crystals are generally euhedral. Later-forming minerals must grow around any pre-existing crystals so tend to have less well-formed crystal shapes.
- If the rock contains both large and small crystals of the same mineral, the large ones grew first.

5. Classify the rock

The mineralogical and textural information you have gathered should allow you to classify the rock by selecting an appropriate classification scheme. Usually, igneous rocks are named on the basis of the modal proportions of the major minerals (i.e. *essential* and *type* minerals).

6. Interpret the sequence of events that formed the rock

The interpretation should be supported by your observations and might address questions such as: Is the rock intrusive or extrusive? Which crystals grew slowly, and which grew rapidly? Which crystals grew first? What igneous environment could produce crystals with these shapes, sizes or textures? What tectonic environment might this rock have formed in (e.g., mid-ocean ridge, subduction zone volcano)? You may wish to include diagrams or cartoons of the relevant geological processes.

Classification and nomenclature of igneous rocks

Igneous rocks can be classified according to their chemical composition, or by their mineralogy. The choice of classification scheme will depend on the types of information available. If only thin sections or hand specimens are available, a petrological scheme must be used. Compositional classification schemes such as TAS require a geochemical analysis of the rock.

Qualitative petrological classification

The scheme below uses essential and type minerals to classify igneous rocks. It can be used in the lab or in the field.

Gill (2010) Table 1.1 Root names and essential minerals for the petrographic classification of fine-grained igneous rocks. For coarse-grained rocks, use the equivalent coarse-grained root name (e.g., gabbro or dolerite instead of basalt). Where percentages are specified, they refer to the volume % based on visual assessment of a thin section. LCP is low-Ca pyroxene, including enstatite and pigeonite.

Root name	Essential minerals required	Possible type minerals
basalt	augite + calcic plag	olivine, LCP (→ tholeiitic basalt[1]), minor nepheline (+ olivine → alkali basalt[1])
basanite	augite + calcic plag + foid (>10%) + olivine (>10%)	nepheline, leucite or analcite (according to the dominant foid)
tephrite	augite + calcic plag + foid (>10%) (olivine < 10%)	nepheline, leucite or analcite (according to the dominant foid)
nephelinite	augite + nepheline	olivine, melilite
leucitite	augite + leucite	olivine, melilite
trachybasalt	Calcic plag + augite + alkali feldspar *or* foid	
andesite	sodic plag[2] + a mafic mineral (pyroxene *or* hornblende *or* biotite)	pyroxene, hornblende or biotite (according to the dominant mafic mineral present)
latite	sodic plag + alkali feldspar	quartz (<20%)
trachyte	alkali feldspar[3] ± sodic plag	quartz (<20%), foid (<10%), aegirine-augite, biotite
phonolite	alkali feldspar[3] ± sodic plag + foid (>10%)	aegirine, riebeckite, biotite; It is also normal to specify the dominant foid if not nepheline, e.g. 'leucite phonolite'
dacite	sodic plag[4] + alkali feldspar + quartz (>20%)	hornblende, biotite
rhyolite[5]	alkali feldspar[3] ± sodic plag + quartz (>20%)	biotite, aegirine

[1] These types of basalt are usually distinguished according to chemical (rather than mineralogical) criteria.
[2] Plagioclase phenocrysts, abundant in most andesites, are commonly zoned with calcic cores; the *mean* feldspar composition is An <50.
[3] Alkali feldspar exceeds plagioclase
[4] Plagioclase exceeds alkali feldspar.
[5] Rhyolites commonly have a glassy groundmass.

How to use this scheme:

1. Identify the minerals in the rock.
2. Assign the root name based on the essential minerals and the grain size. (NB the root names in the table apply only to fine-grained rocks.)
3. Add qualifiers based on the type minerals present.

Essential minerals are implied by the root name. Example: the rock name “granite” tells you that the rock contains quartz, plagioclase and alkali feldspar.

Type minerals are not implied by the root name. Example: the rock name “granite” doesn’t tell you whether there are any minerals *in addition to* quartz, plagioclase and alkali feldspar. The rock name is therefore used to describe the type minerals. If there is more than one type mineral, they are listed with the least abundant first. Example: biotite muscovite granite.

Essential and type minerals together constitute the major minerals. Minerals that are not major minerals are accessory or secondary.

- Accessory minerals constitute only a small proportion of the rock (usually less than 5%). They are not important for classifying the rock type, composition and origin. They are not included in the rock name. Common examples: opaque oxides; apatite.
- Secondary minerals form after the magma had completely crystallised. They may fill void spaces or replace original minerals. They are not included in the rock name. Common examples: serpentite (alteration of olivine); chorite (alteration of biotite); zeolites (filling amygdales).

Some additional pointers:

- For andesites and diorites, the mafic mineral is often included in the rock name. Example: hornblende andesite.
- Diorite can usually be distinguished from gabbro by its abundant hornblende.
- If a mineral occurs as phenocrysts in a fine-grained rock, it can be included in the rock name by adding “XX-phyric” to the root name. This nomenclature can be used even if the phenocryst plase is one of the essential minerals. Example: plagioclase-phyric basalt.

Fine-grained or volcanic rock	**Medium or coarse-grained equivalent**	**Common phenocryst minerals** (this list is neither exhaustive nor prescriptive)
	Felsic rocks	
rhyolite	granite	quartz, alkali feldspar
dacite	granodiorite	quartz, Na-rich plagioclase, alkali feldspar
trachyte	syenite	alkali feldspar
phonolite	nepheline syenite	alkali feldspar, nepheline or leucite, aegirine/augite
	Mafic rocks	
andesite	diorite	Na-rich plagioclase, pyroxene or amphibole
basalt	gabbro	Ca-rich plagioclase, pyroxene, olivine

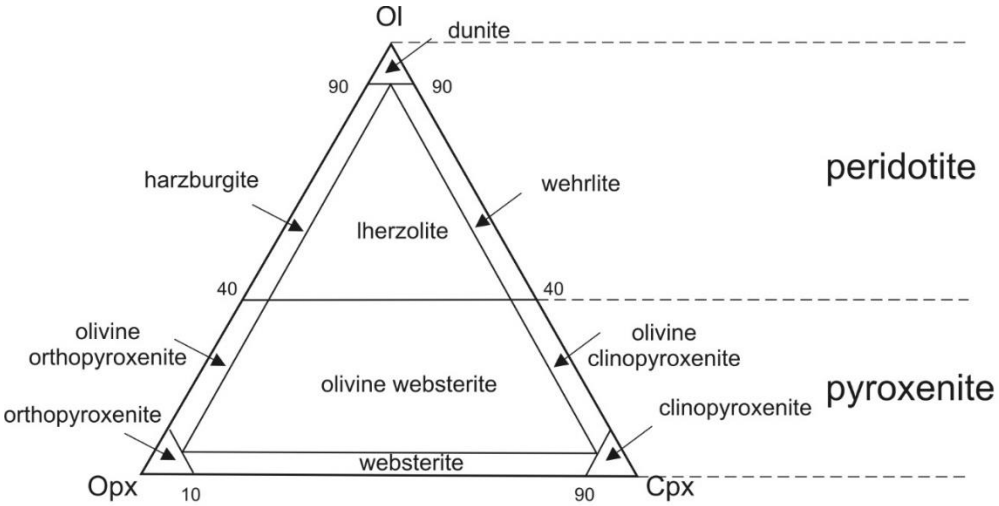
Fine-grained rock types, their coarse-grained equivalents, and common phenocryst minerals

Quantitative petrological classification

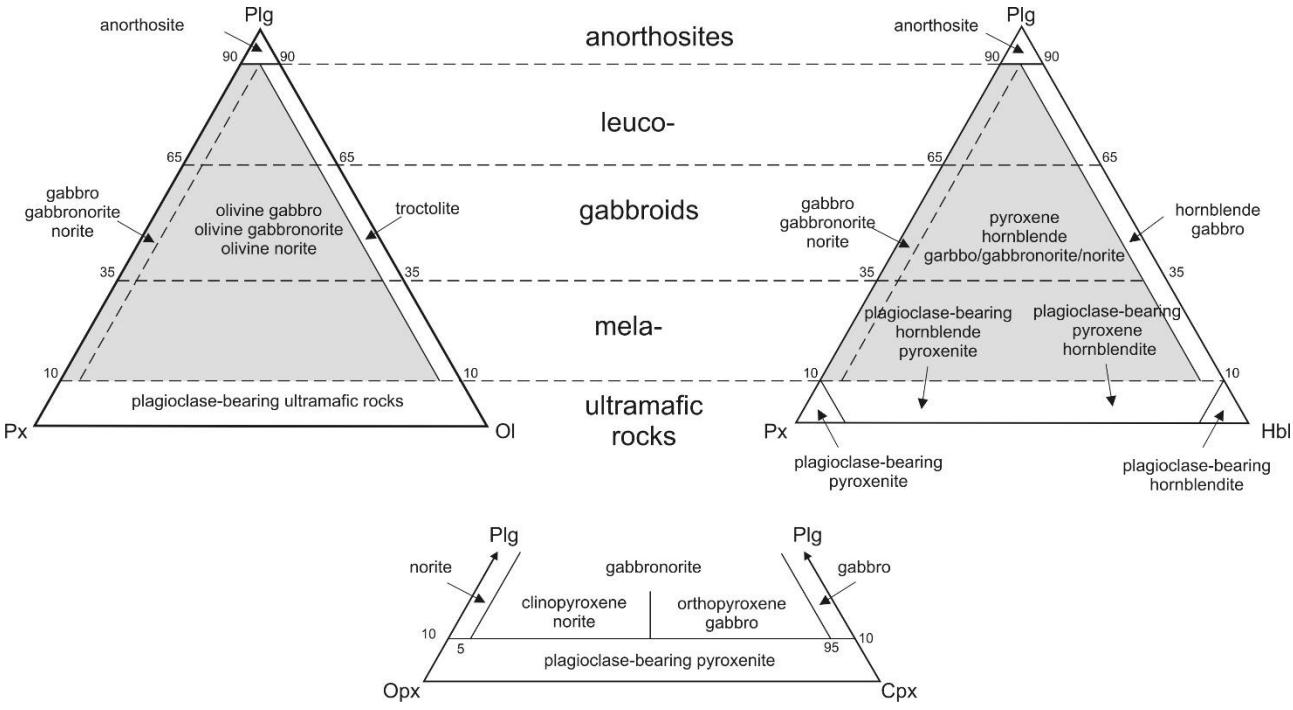
These schemes achieve more precise classification of the rock. They require quantitative estimates or measurements of the relative modal abundance of key minerals.

Ultramafic and mafic rocks

Nomenclature of crystalline ultramafic rocks (>90% mafic minerals) based on olivine (ol), orthopyroxene (opx) and clinopyroxene (cpx) contents (Streckeisen, 1973):



Nomenclature of crystalline mafic rocks based on plagioclase (plg), total pyroxene (px), olivine (ol), clinopyroxene (cpx), orthopyroxene (opx) and hornblende (hbl) contents (Streckeisen, 1967). Rocks projected within the shaded fields can be further classified using the bottom diagram.



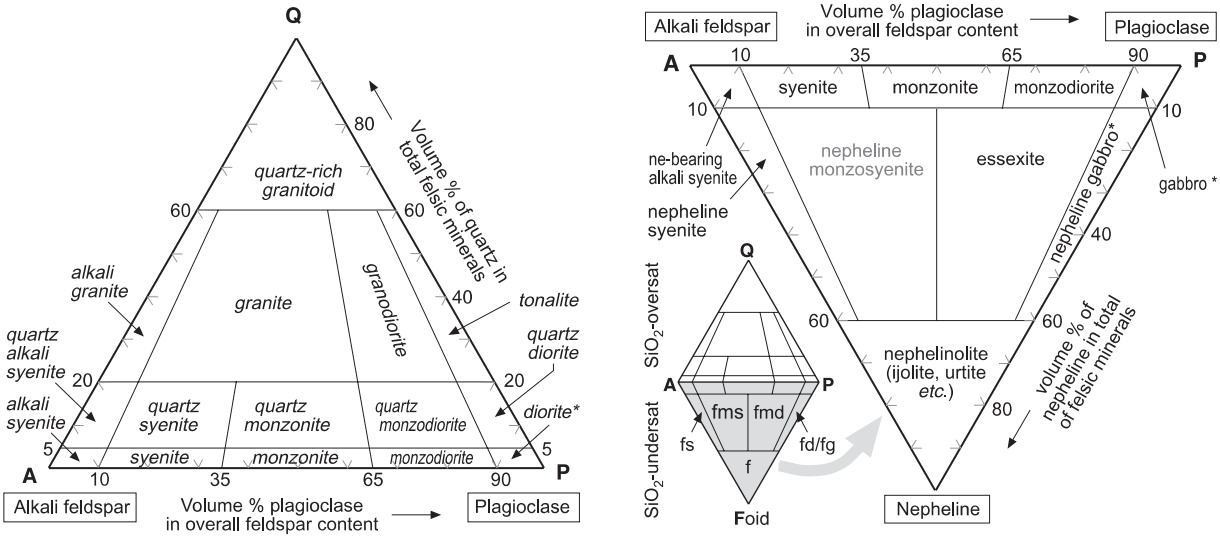
Granites and granitoids

Quartz-Alkali feldspar-Plagioclase-Feldsathoid (QAPF) classification. Used for silica-saturated rocks containing the essential minerals Quartz-Alkali feldspar-Plagioclase (QAP), and for silica-undersaturated rocks containing the essential minerals Alkali feldspar-Plagioclase-Feldspathoid (APF). A rock *cannot* contain both quartz and feldspathoid. In EART 22201 we will not study rocks that contain feldspathoid minerals.

To use the QAPF scheme:

1. Determine the modal mineralogy of the sample. Estimate the % of the essential minerals quartz (Q), alkali feldspar (A), plagioclase (P) and feldspathoid (F) in the sample.
2. If there is no feldspathoid, scale the proportions of Q, A and P so that they add up to 100%, and plot the result on the QAP triangle.
3. If there is no quartz, scale the proportions of A, P and F so that they add up to 100%, and plot the result on the APF triangle. The feldspathoid ("foid" for short) in the diagram below is nepheline, but if the rock contains a different feldspathoid (e.g. leucite) then this is substituted for nepheline in the diagram.
4. The QAP(F) classification provides the root name of the rock. Further classification can be achieved by examining the type minerals of the rock.

Plutonic QAPF diagrams: Gill (2010) figures 8.1 and 9.4



The rock names in these diagrams are for medium- and coarse-grained rocks. There are equivalent diagrams showing the rock names for fine-grained silica-rich igneous rocks.

Essential and type minerals in granitic rocks: Gill (2010) Table 8.2

	Diorite	Tonalite	Granodiorite	Granite[¶]	Alkali granite
Essential minerals	• plagioclase • one or more mafic minerals as below	• plagioclase • quartz	• sodic plagioclase • quartz • alkali feldspar *plag > alk fsp*[$]	• alkali feldspar • quartz • sodic plagioclase *alk fsp ≥ plag*	• alkali feldspar • quartz
Type minerals	• quartz • hornblende • biotite • augite	• hornblende • biotite[@]	• hornblende • biotite[@]	• hornblende • biotite[§ @] • muscovite[§] • tourmaline • garnet (almandine) • cordierite • andalusite	• riebeckite or other alkali amphibole • aegirine-augite (These alkaline minerals are discussed in Ch. 9)
Colour index	Melanocratic or mesocratic	Mesocratic*	Mesocratic or leucocratic	Leucocratic	Leucocratic
Common secondary minerals	• chlorite, **uralite** or **iddingsite** replacing pyroxene, hornblende or biotite • **sericite** or epidote replacing feldspars (Quartz is not susceptible to alteration.)				

[$] For the distinction between alkali feldspar and plagioclase in plutonic rocks under the microscope, see Box 8.1. **Perthite**, when present, is included with alkali feldspar; **antiperthite** is counted as plagioclase.
[§] The name 'two-mica granite' is often used for a granite containing *both* biotite and muscovite (Plate 8.2).
* A leucocratic variant of tonalite, consisting essentially of sodic plagioclase and quartz with biotite, is given the name **trondhjemite**.
[¶] The terms (quartz-) monzonite and (quartz-) syenite designate quartz-poor equivalents of sodic and potassic granites (see Fig. 8.1).
[@] Often contains inclusions of radioactive accessory minerals like zircon that give rise to pleochroic haloes (Plate 8.10).

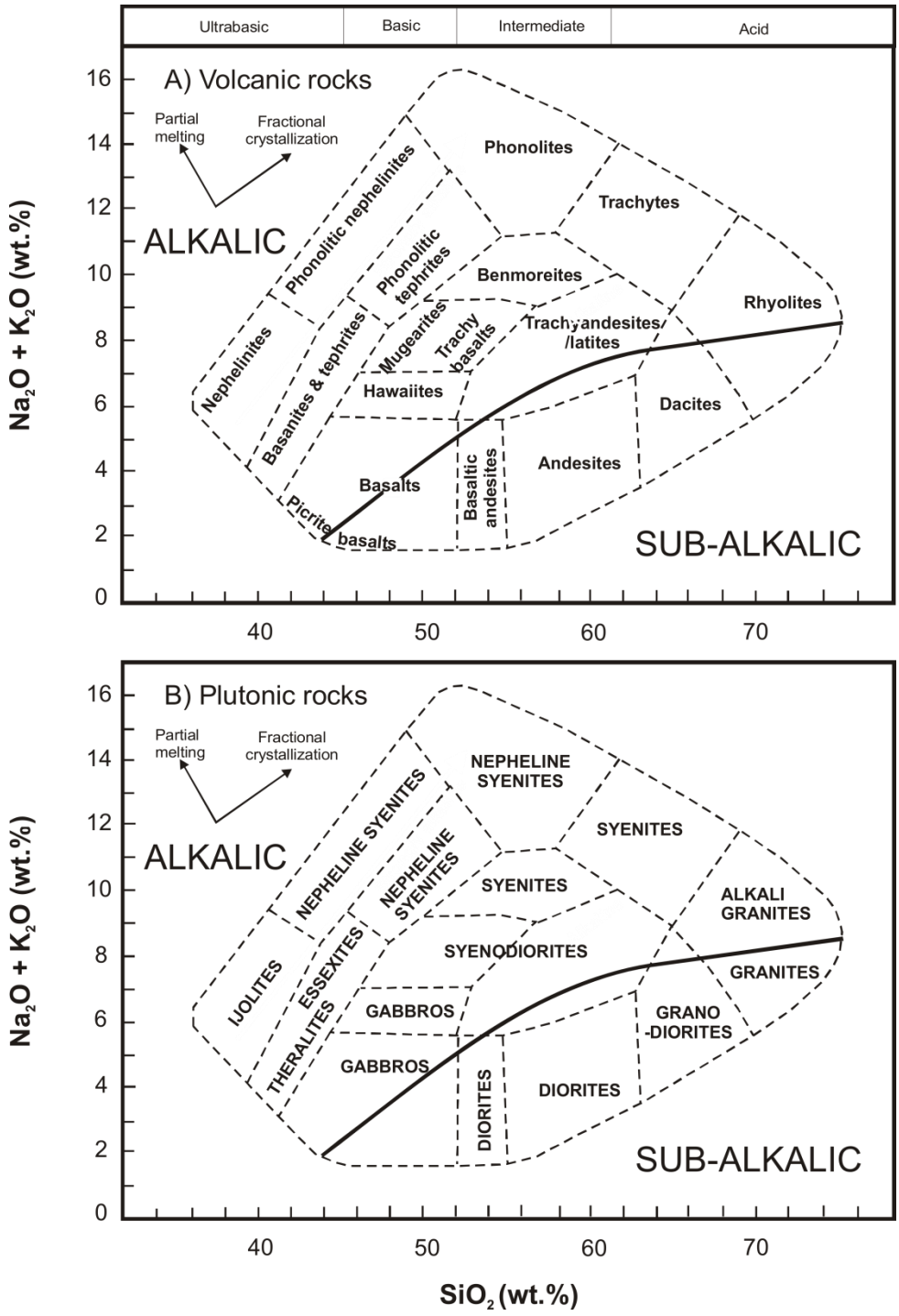
Type minerals in S-type, I-type and A-type granitoids: Gill (2010) Table 8.3.1

	I-type granitoids	S-type granitoids	A-type granitoids
Major element composition	Metaluminous*	Peraluminous*	Metaluminous to peralkaline; High ΣFeO/MgO (see Fig. 8.15)
Normative minerals	Normative diopside	Normative corundum	Normative diopside ± acmite
Range of rock types represented	Plutons commonly include a wide range of rock types from basic to acid (SiO_2 56%–77%)	Generally confined to high-SiO_2 leucogranite (SiO_2 64%–77%). No associated mafic rocks.	Typically high-SiO_2 granitoids (often associated with syenite)
Type minerals	Hornblende and biotite	Muscovite (usually with biotite) ± other Al-rich minerals (Box 8.2)	Mafics are Fe-rich biotite or alkali pyroxene/amphibole
Range of $(^{87}Sr/^{86}Sr)_0$	0.704–0.706	0.708–0.765	0.702–0.717[§]
Xenolith lithologies	Mafic hornblende-bearing xenoliths of igneous appearance	Metasedimentary xenoliths	Various
Associated economic deposits	Porphyry Cu, Mo sulphides ± Ag-Au-bearing pyrite veins	Sn, W , U (Li, Be, B) (see Fig. 8.3.1)	Zr, Hf, Nb, Ta, Y, **REE**, U, Th (**HFS** elements – see Fig. 2.7.1)

* Chappell and White (1974) placed the I-type/S-type boundary at $Al_2O_3/(Na_2O+K_2O+CaO)_{mol} = 1.1$ rather than 1.0.
[§] The range for the White Mountain magma series of New Hampshire is an exception, extending up to 0.736 (Eby, 1990).

Chemical classification of igneous rocks

Total alkali vs. silica (TAS) plot for volcanic and plutonic systems:



SiO_2 content (wt.%)	Classification
<45	ultramafic
45-52	mafic
53-62	intermediate
>63	felsic

Common igneous terms used in rock descriptions

1. Degree of crystallinity

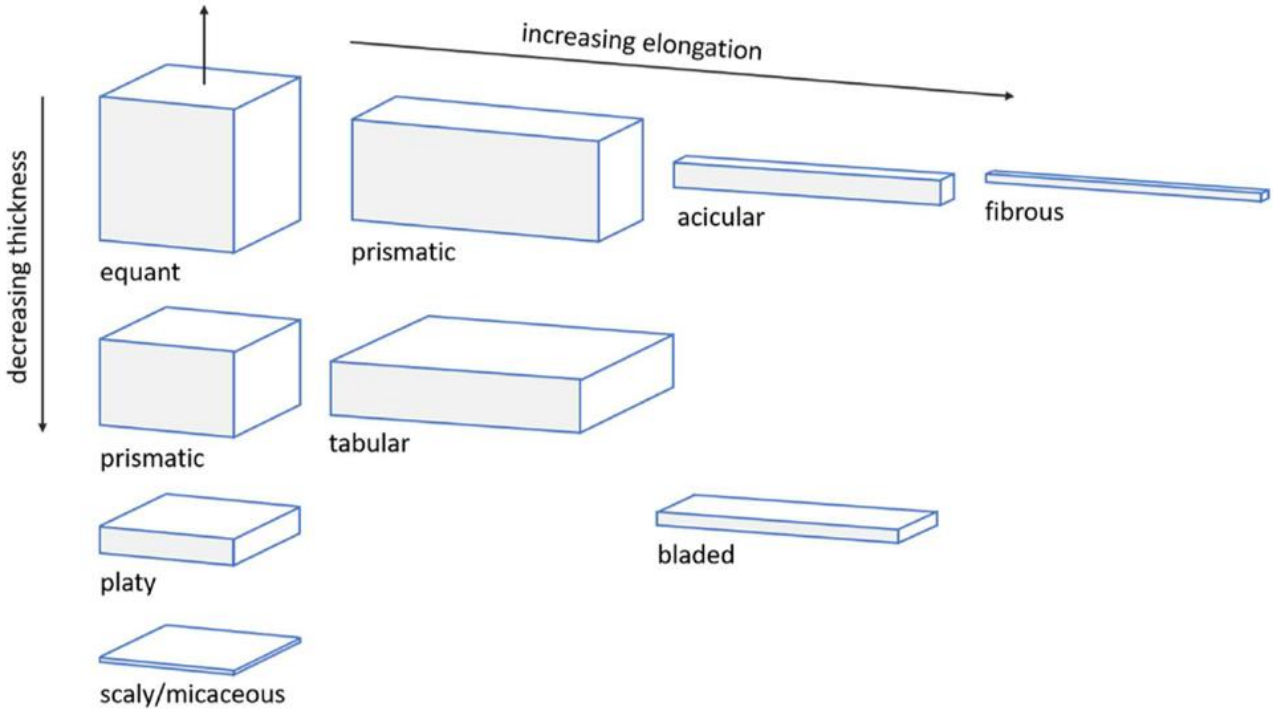
holocrystalline	The rock is entirely composed of crystals.
hypocrystalline	The rock consists partly of crystals and partly of a glassy base.
holohyaline	The rock consists entirely of glass. The terms glassy and vitric are also used.

2. Sizes of individual crystals

aphanitic	None of the crystals (except phenocrysts) can be seen by the naked eye. Rock cooled quickly so apply volcanic terminology.
phaneritic	All the principal minerals are visible to the naked eye. Rock cooled slowly so apply plutonic terminology.
groundmass	A fine-grained crystalline matrix in which larger crystals may be embedded.
microcrystalline	Groundmass crystals can be distinguished under the microscope.
cryptocrystalline	Groundmass crystals cannot be distinguished under the microscope.
equigranular	Crystals are all approximately equal in size.
inequigranular	Crystals have a range of different sizes.

Textural terms should normally be accompanied by estimates or measurements of the crystal size.

3. Crystal shape and habit



euhedral	Shows perfect or near-perfect crystal form. Typical of early-crystallizing minerals.
subhedral	Some crystal faces are moderately well developed.
anhedral	Crystal faces are absent; has an irregular or rounded shape.
skeletal	Shows a branched, tree-like form with internal cavities, often with an overall euhedral outline. Forms under very rapid growth conditions.
embayed	Crystal faces are dimpled, caused by partial resorption or dissolution of the mineral.
sieve	Abundant small and interconnected glass inclusions found inside the crystal, giving it a spongy or porous appearance. Usually seen in feldspars in volcanic rocks, e.g. sieve-textured plagioclase.

4. Textural relationships of crystals

Equigranular textures

granular	Most of the grains are approximately equant.
granitic	Medium- to coarse-grained rocks in which all or most of the minerals are equal in size and display interlocking borders. Most of the grains are anhedral or subhedral. Generally used for rocks of granitic composition.
microgranitic	Most of the grains are anhedral or subhedral, and <5 mm diameter.

Inequigranular textures

porphyritic	Two distinct sizes of crystals. Large phenocrysts are distributed through a fine-grained groundmass. If the phenocrysts are set in glass, the term vitrophyric can be used.
glomeroporphyritic	Phenocrysts have drifted together and attached to form clusters called glomerocrysts. A common texture in andesites. The process that forms this texture is called synneusis.
aphyric	Not porphyritic; does not contain phenocrysts. Most commonly used to describe volcanic rocks, since the lack of phenocrysts is worth commenting on. This may indicate that the magma spent little or no time in a large magma chamber.
seriate	A texture where the crystals show a wide and continuous range of sizes. Can be difficult to identify in a 2D thin section.
poikilitic	Large crystals (oikocrysts) completely enclose smaller grains (chadacrysts) of a different mineral.
ophitic	A specific case of poikilitic texture where plagioclase laths are enclosed by larger oikocrysts of augite. Typical in dolerites.
subophitic	Similar to ophitic texture, but where plagioclase laths are only partly surrounded by larger oikocrysts of augite.

interstitial	An anhedral mineral that has crystallized at a late stage and filled the pore spaces (interstices) between earlier crystals.

Oriented and aligned textures

flow banded	Irregular lines or bands of differing colour and/or texture and/or composition, produced during the flow of highly viscous liquids such as andesitic to rhyolitic lavas.
trachytic	A variant of microlitic texture, where groundmass feldspar crystals show a well-developed preferred orientation in the direction of lava flow. Typically seen in trachyte, trachybasalt and basalt lavas.
eutaxitic	The banded appearance of welded pyroclastic rocks, caused by the collapse of pumice shards to form fiammé.
fiammé	Collapsed pumice fragments, usually with a large aspect ratio, in welded tephra deposits. The texture has a flame-like appearance.
cumulate	Formed by crystals settling through a magma, usually gravity settling. Most of the liquid is expelled as the cumulate compacts. The rock will have a framework of larger, closely packed cumulus crystals, which may have a preferred alignment producing a layered fabric. The interstitial spaces between the cumulus crystals are filled by intercumulus minerals.

Exsolution textures

perthitic	An alkali feldspar that has exsolved in to Na-rich and K-rich domains. The exsolution lamellae are expressed as strings, patches or lenses. If the strings are very small, use microperthitic.

Zoning textures

normal, reverse	These terms refer to solid-solution composition changes from the core to the rim of the crystal. In normally zoned crystals, the core has the high-T crystal composition and the rim has the low-T composition. In reverse-zoned crystals, the opposite is true.
oscillatory	Most common in plagioclase feldspars, but can also be observed in alkali feldspars, clinopyroxenes and amphiboles. Most common in volcanic rocks from subduction zones.

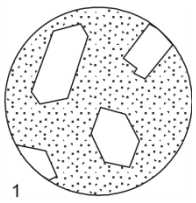
Textures formed by cavities or holes

vesicular	A volcanic rock that contained gas bubbles. Vesicles may be rounded, elongate or irregular shapes.
amygdaloidal	A texture formed when vesicles are filled by secondary minerals such as zeolites. The filled vesicles are called amygdales.

Some common igneous textures

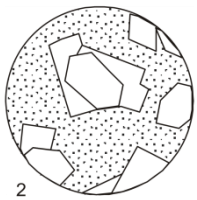
Volcanic rocks

Lavas



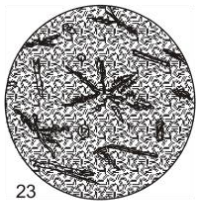
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Porphyritic



2

Glomeroporphyritic



23

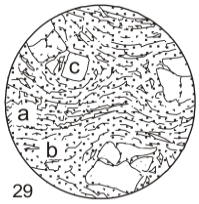
Crystallite and microlites in glass



22

Trachytic texture

Ignimbrites



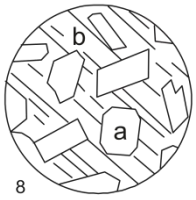
29

Welded ignimbrite, with flattened and welded pumice (a), shards (b) and broken crystals (c)

Eutaxitic texture

Plutonic rocks

Gabbros and dolerites



8

Poikilitic texture - many crystals of 'a' enclosed by one large grain of 'b'



9

Ophitic texture - plagioclase laths enclosed by pyroxene



10

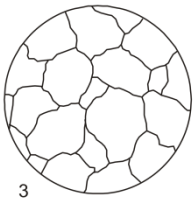
Interstitial texture



20

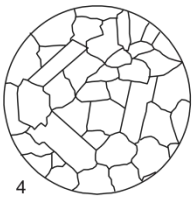
Banding in a cumulate rock. Plagioclases in lowest layer shows preferred orientation

Granites and granitoids



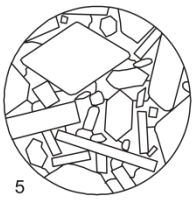
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Allotriomorphic granular



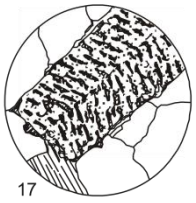
4

Hypidiomorphic granular



5

Idiomorphic granular



17

Perthitic texture

In alkali feldspars

How to distinguish between olivine and augite

Olivine and augite (clinopyroxene) are often found together in basalts. They have similar relief and interference colours and are sometimes difficult to tell apart. Here are some key features to look for in distinguishing between them. Those in italics are most diagnostic.

Feature	Olivine	Augite
Colour	Colourless in PPL	May be pale *pink to brown* or grey in PPL
Fracture and cleavage	*Irregular curved fractures*, often with alteration. Cleavage rarely visible	One cleavage in elongate sections, and *two perpendicular cleavages* in basal sections
Birefringence	0.035-0.052: up to third-order colours	0.018-0.033: up to second-order colours
Extinction angle	Euhedral crystals have straight extinction relative to crystal faces	*Inclined extinction* relative to cleavage
Alteration	*Susceptible to alteration* to serpentine (colourless) or iddingsite (red-brown)	More resistant to alteration

How to distinguish between orthopyroxene and clinopyroxene

- Extinction angle is the most widely used means of distinguishing between orthorhombic and monoclinic pyroxenes. To use this method, take care to select a crystal that shows a single cleavage trace (not two cleavages), and that shows the highest available interference colour.

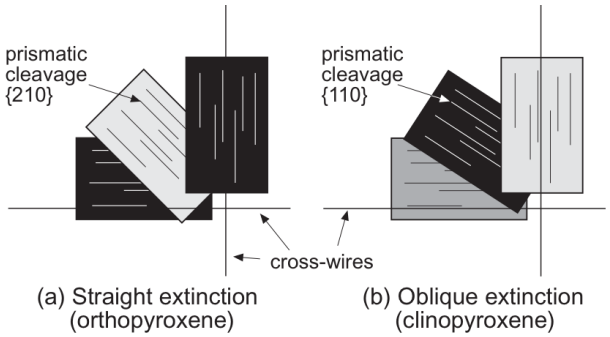
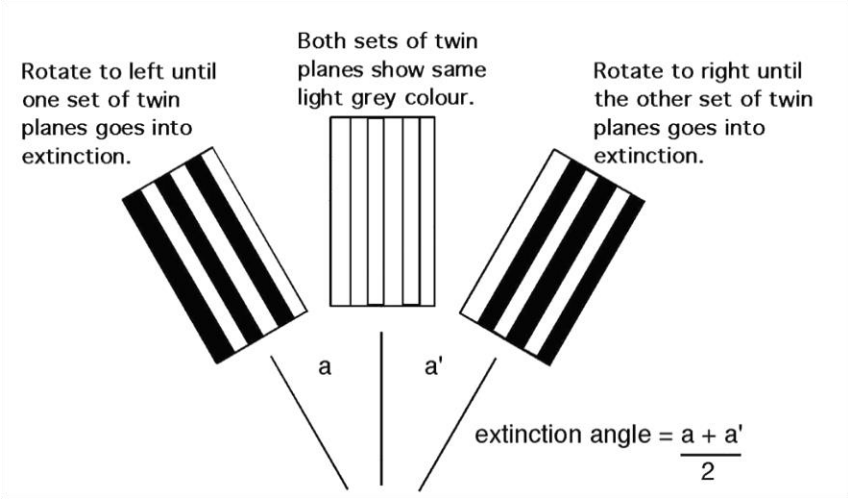


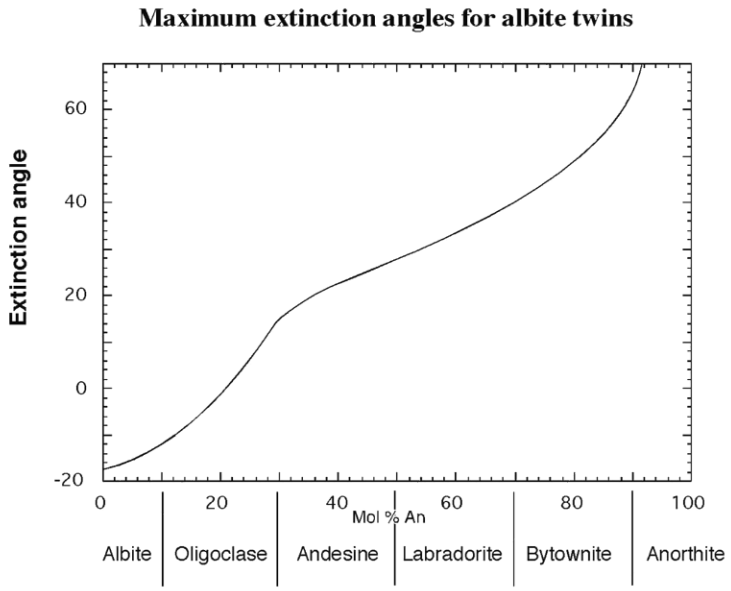
Fig. 4.2.1 The distinction in crossed polars between (a) straight or *parallel* extinction in an α–γ (100) section of an *orthopyroxene* such as enstatite, and (b) *inclined* extinction in an α–γ (010) section of a *clinopyroxene* such as augite.

- Colour and pleochroism: Clinopyroxenes may have a pink to brown colour, and some may show pleochroism. Orthopyroxene is usually colourless and non-pleochroic. The only exception is enstatite ($Mg_2Si_2O_6$ orthopyroxene) which sometimes exhibits a weak but very distinctive pale pink to pale green pleochroism.
- Interference colours. Orthopyroxenes have low first-order interference colours. Most clinopyroxenes have higher interference colours – typically second-order.
- Twinning. Orthopyroxenes never exhibit twinning, whereas clinopyroxenes may do so.

Procedure for determining plagioclase composition

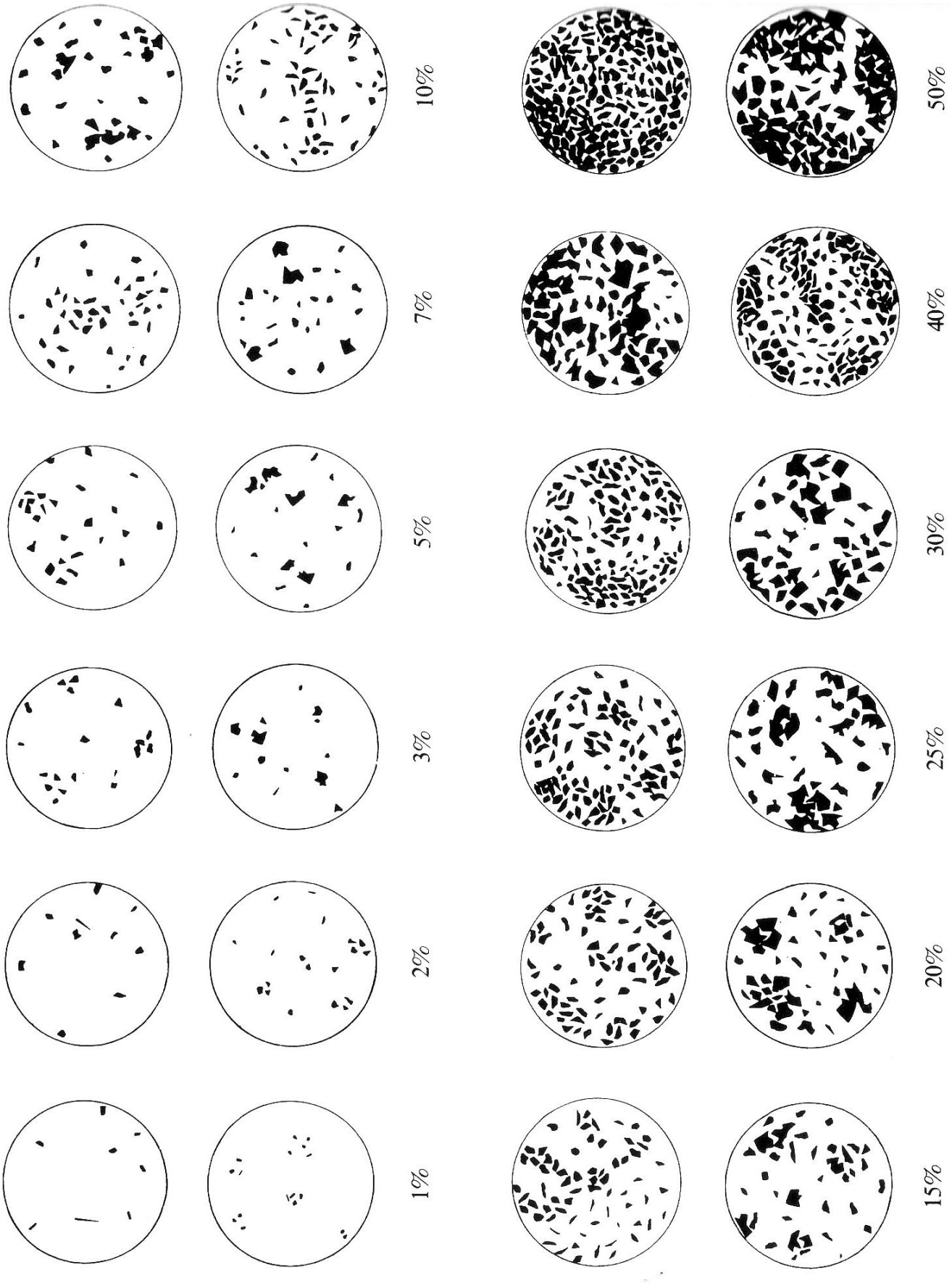


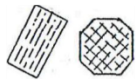
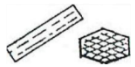
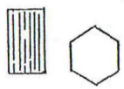
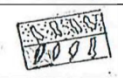
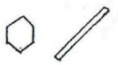
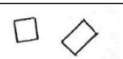
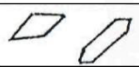
1. Look for a crystal which, when oriented with the twin planes N-S, both sets of twins are the same shade of light grey throughout the entire crystal. Crystals in this orientation should have symmetric extinction angles, i.e. the angles a and a' (on the diagram above) should be within a few degrees of each other.
2. Measure angles a and a' by rotating the stage.
3. If a and a' are *not* within a few degrees of each other, then the crystal is not suitable and you should discard this reading. Search for another crystal.
4. If a and a' *are* within a few degrees of each other, then calculate extinction angle as $\frac{a+a'}{2}$.
5. Measure a few more crystals.
6. Your measurements will vary because different crystals are in different orientations. Use your highest measured extinction angle to determine the composition.
7. Plot your highest value on the graph below to obtain the plagioclase composition.



The curve shows how extinction angle varies with plagioclase composition. Note that you cannot distinguish negative extinction angles from positive extinction angles, so an angle of 15° could be -15° (= albite) or +15° (= oligoclase). Another optical method is required to tell these apart, e.g. measuring their refractive index (Becke Line Test).

Chart for estimating modal proportions of minerals in rocks



	CRYSTAL SYSTEM	SHAPE (Ideal)	CLEAVAGE	RELIEF	COLOUR (Natural)	INTERFERENCE COLOUR	TWINNING	OTHER FEATURES
OLIVINE	Orthorhombic		One (poor)	high	colourless	intermediate	-	Irregular fractures, often associated with alteration to green or brown fibrous SERPENTINE (+opaque Fe oxide).
ORTHOPYROXENE	Orthorhombic		Two (90°)	high	colourless or very pale grey	low	-	Most sections show 1 cleavage and straight extinction. Alteration (as for olivine). May show unmixing of augite.
AUGITE (CPX)	Monoclinic			high	pale grey-brown	intermediate	occasional (simple or multiple)	Most sections show 1 cleavage and oblique extinction.
HORNBLENDE	Monoclinic		Two (60/120°)	high	green/brown PLEOCHROIC	intermediate	occasional (simple or multiple)	Most sections show 1 cleavage and oblique extinction. May be altered to CHLORITE (pale green, low interference colours).
BIOTITE	Monoclinic		One	mod.	brown PLEOCHROIC	intermediate	-	Sections showing cleavage have straight extinction. Speckled appearance close to extinction. Alteration to chlorite.
MUSCOVITE	Monoclinic			mod.	colourless	intermediate	-	Sections showing cleavage have straight extinction. Speckled appearance close to extinction.
QUARTZ	Trigonal		-	low	colourless	low	-	Always unaltered. May show patchy extinction (due to slight deformation).
FELDSPAR (general)	Monoclinic or Triclinic		Two (~90°)	low	colourless	low	common	Typically some degree of alteration to clay minerals (fine particles) and small flakes of secondary muscovite (sericite).
Plagioclase Feldspar (Triclinic)			"	"	"	"	multiple (mainly 1direction) ±simple (same direction)	Alteration to sericite, but may also produce small grains of EPIDOTE (colourless or greenish, bright interference colours). 
K-feldspar	Sanidine (Monoclinic)		"	"	"	"	simple	Often unaltered (may be confused with quartz) Look for twinning and cleavage. 
	Orthoclase (Monoclinic)		"	"	"	"	simple	Alteration to clay mins. Tend to show some degree of unmixing of Na-rich feldspar ORTHOCLASE PERTHITE 
	Microcline (Triclinic)		"	"	"	"	multiple in 2 directions (~90°) on fine scale	Alteration to clay mins. Tend to show some degree of unmixing of Na-rich feldspar MICROCLINE PERTHITE 
GARNET	Cubic		-	high	colourless (grey) or pale brown	isotropic	-	**COMMON OPAQUE MINERALS**
CALCITE	Trigonal		Three (not 90°)	low	colourless	high	multiple	**HEMATITE** (Fe_2O_3) – may be deep red in very thin flakes
APATITE $Ca_5(PO_4)_3$	Hexagonal		-	mod.	colourless	low	-	**MAGNETITE** ($FeO.Fe_2O_3$)
ZIRCON $ZrSiO_4$	Tetragonal		-	v. high	colourless (grey)	intermediate	-	**CHROMITE** ($FeO.Cr_2O_3$) } tend to form equant grains (cubic system]
SPHENE $CaTiSiO_5$	Monoclinic		Two	v. high	pale brown	high	simple	**ILMENITE** ($FeO.TiO_2$) – tends to form elongate crystals